

UMAP Visualization of GPT-5 Reddit Discussion Clusters (UMAP–Nomic + HDBSCAN)

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Abstract—This study presents a comprehensive quantitative analysis of user sentiment following the launch of GPT-5, focusing specifically on discussions within the official OpenAI AMA Reddit thread. The primary goal was to systematically identify and categorize user pain points and concerns using advanced natural language processing (NLP) techniques. My analysis processed 1,972 unique Reddit posts and comments using two state-of-the-art embedding models: BAAI/bge-base-en-v1.5 and nomic-ai/nomic-embed-text-v1.5. Through a modular experimental design, evaluating various combinations of Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) for dimensionality reduction, and KMeans, DBSCAN, and HDBSCAN for clustering, I identified the optimal configuration as UMAP (nomic-embed-text-v1.5) combined with HDBSCAN. This configuration achieved a high silhouette score of 0.603 across 21 distinct clusters. Subsequent topic modeling, enhanced by Maximal Marginal Relevance (MMR) for interpretability, revealed several critical pain points. The central concerns include the removal of beloved legacy models (particularly GPT-4o), a perceived downgrade in creative writing and conversational personality, and restrictive usage limits/pricing that reduced the perceived value for paid subscribers. These findings provide vital data-driven insights into the user experience challenges associated with major large language model (LLM) transitions.

Index Terms—Text Clustering, Topic Modeling, UMAP, HDBSCAN, BERTopic, User Sentiment Analysis, Large Language Models

I. INTRODUCTION

The launch of GPT-5 by OpenAI marked a significant, yet complex, milestone in the evolution of large language models (LLMs). While highly anticipated for its advancements in reasoning and efficiency, the rollout was accompanied by substantial user friction and widespread complaints across community platforms, particularly concerning the simultaneous deprecation of popular legacy models. Unlike previous releases, which often saw universally positive reception, GPT-5’s introduction highlighted a critical divergence between technical benchmark improvements and real-world user satisfaction.

This research addresses the need to move beyond anecdotal feedback by systematically quantifying and categorizing the user experience during this major model transition. By applying computational text analysis to a high-volume, focused discussion thread—the official GPT-5 AMA on Reddit—this study aims to provide an objective, data-driven understanding of how users perceive model changes, where frustration is

concentrated, and the specific functional and emotional pain points that emerged.

A. Research Objectives

The core objectives of this study are threefold:

- 1) **Evaluate State-of-the-Art Embedding and Clustering Methods:** To perform a rigorous comparison of modern text embedding, dimensionality reduction (PCA, UMAP), and clustering techniques (KMeans, DBSCAN, HDBSCAN) to determine the optimal configuration for semantically grouping highly subjective user feedback.
- 2) **Identify and Quantify User Pain Points:** To systematically cluster the Reddit discussions and quantify the sentiment (via mean comment score) and prevalence (via cluster size) of distinct user complaints, thereby moving beyond simple sentiment analysis.
- 3) **Extract and Label Key Themes:** To apply advanced topic modeling, including representation re-ranking and generative labeling, to derive clear, human-interpretable themes and topics that accurately describe the major challenges users faced with the GPT-5 transition.

B. Contribution

This study makes several key contributions to the field of computational social science and human-AI interaction:

- **Empirical Validation of LLM Transition Friction:** Provides concrete evidence and categorized data detailing the negative user reception, emphasizing that emotional and preference-based factors (e.g., model personality) are as critical as technical performance.
- **Methodological Framework:** Establishes and validates a robust, modular NLP methodology—specifically leveraging UMAP + HDBSCAN with high-performance embeddings—that is highly effective for structuring noisy, high-volume user-generated content from platforms like Reddit.
- **Actionable Insights for Deployment Strategy:** Offers specific, actionable findings regarding the critical impact of removing legacy models and altering pricing/usage parity, which can directly inform future LLM deployment, communication, and model retention policies.

II. LITERATURE REVIEW AND BACKGROUND

A. Contextual Embeddings and Dimensionality Reduction

The foundation of modern text analysis rests on contextual embeddings, which overcome the limitations of early methods by generating high-dimensional vectors that encode semantic relationships. This study utilized models selected from the MTEB (Massive Text Embedding Benchmark) leaderboard, including BAAI/bge-base-en-v1.5 for its strong general performance and nomic-ai/nomic-embed-text-v1.5 for its robust capabilities in handling long-context and retrieval tasks, reflecting the nature of deep Reddit discussions.

Before clustering, dimensionality reduction is critical for mitigating the curse of dimensionality. The study employed two distinct techniques:

- **Principal Component Analysis (PCA):** A linear technique used to reduce the dimensionality while maximizing variance preservation. This provides a baseline, computationally efficient representation.
- **Uniform Manifold Approximation and Projection (UMAP):** A non-linear technique crucial for capturing the manifold structure of the data. UMAP often yields superior clustering results for high-dimensional text embeddings by preserving local data relationships.

B. Clustering Algorithms for Semantic Grouping

A thorough evaluation of the underlying data structure requires testing multiple clustering algorithms, as each makes different assumptions about the nature of the data:

- **KMeans:** Chosen for its computational efficiency and simplicity, it partitions data into a pre-defined number of clusters (set to $k = 8$ in our preliminary tests) by minimizing inertia. However, it assumes spherical, equally-sized clusters.
- **DBSCAN (Density-Based Spatial Clustering of Applications with Noise):** A density-based algorithm that identifies clusters of arbitrary shapes and is capable of explicitly flagging outliers (noise points). This approach does not require specifying the number of clusters in advance.
- **HDBSCAN (Hierarchical DBSCAN):** An extension of DBSCAN that builds a cluster hierarchy and extracts the most stable clusters. This method is particularly well-suited for text data, where clusters often exhibit varying densities and sizes, and proved to be the optimal choice in our final configuration.

C. Modern Topic Modeling with BERTopic

To move from raw semantic clusters to interpretable themes, this study utilized **BERTopic**. This framework combines the strengths of transformer-based embeddings and HDBSCAN clustering with a class-based TF-IDF (c-TF-IDF) representation layer.

- **c-TF-IDF:** This technique generates sparse vector representations for each cluster, allowing for the extraction of keywords that are highly representative of a specific

topic within the corpus, as opposed to keywords common across all topics.

- **Representation Modeling:** To enhance the quality and diversity of topic keywords, I applied representation re-ranking using **Maximal Marginal Relevance (MMR)**. MMR selects keywords that are both highly similar to the central topic representation and dissimilar from one another, ensuring a descriptive and varied set of terms.

III. METHODOLOGY

The methodology employed a rigorous, multi-stage computational pipeline to transform unstructured Reddit data into structured, actionable insights, enabling the objective analysis of user pain points.

A. Data Collection and Preprocessing

Data was collected from the official GPT-5 AMA thread on Reddit using the AsyncPRAW library. The target dataset included the main post and all associated comments, totaling **1,972** unique entries after initial filtering of duplicates and system messages.

- 1) **Preprocessing Pipeline:** A custom pipeline was applied to clean the noisy social media data. This involved systematically removing URLs, user mentions (@username), subreddit references (r/subreddit), and markdown formatting (**text**, [text](url)).
- 2) **Noise Reduction:** The final dataset was filtered to retain only texts with sufficient content length (minimum 10 characters) to ensure that the embeddings were based on meaningful user input, culminating in a clean corpus of 1,972 entries.

B. Embedding Model Selection and Implementation

To prepare the text for clustering, two distinct high-fidelity embedding models were used, reflecting a comparative approach in the experimental design. Embeddings were generated as 768-dimensional dense vectors using a **batch size of 32** to optimize GPU memory utilization and processing speed.

1) Embedding Models:

- BAAI/bge-base-en-v1.5
- nomic-ai/nomic-embed-text-v1.5

- 2) **Dimensionality Reduction:** Each set of 768-dimensional embeddings was reduced to 50 dimensions using two complementary techniques for comparative evaluation:

- **PCA (Principal Component Analysis):** For linear variance preservation.
- **UMAP (Uniform Manifold Approximation and Projection):** For non-linear preservation of local structure, configured with `min_dist=0.0` and `metric='cosine'` for optimal semantic separation.

C. Clustering Experimentation and Selection

A modular experimental design was executed, evaluating all 12 combinations of the two embedding models, two dimensionality reduction techniques, and three clustering algorithms.

1) Clustering Algorithms:

- **KMeans:** Configured with $k = 8$ clusters.
- **DBSCAN:** Configured with $\epsilon = 0.5$ and $\text{min_samples}=5$.
- **HDBSCAN:** Configured with $\text{min_cluster_size}=15$ and $\text{cluster_selection_method}='eom'$.

2) **Optimal Configuration Selection:** Each combination was rigorously evaluated using quantitative metrics, including the **Silhouette Score** for cohesion and the **Calinski-Harabasz Index** for separation. The superior configuration chosen for final topic modeling was: **UMAP (nomic-embed-text-v1.5) + HDBSCAN**, which yielded the highest Silhouette Score of **0.603** across **21** distinct clusters. HDBSCAN was configured to identify clusters with a minimum size of 15, ensuring the stability and relevance of the final topics.

D. Topic Modeling and Interpretation

The final clustering results were fed into the BERTopic framework for topic extraction and labeling.

- 1) **Topic Extraction:** The HDBSCAN cluster assignments were used by BERTopic to generate topics, with an outlier topic (−1) containing 759 posts identified and separated from the main thematic clusters.
- 2) **Interpretability Enhancement:**
 - **Representation Re-ranking (MMR):** Maximal Marginal Relevance was applied to the c-TF-IDF vectors to select more diverse and representative keywords, significantly improving interpretability.
 - **Generative Labeling (Flan-T5):** The `google/flan-t5-small` model was employed to generate concise, descriptive labels for the identified topics based on sample documents and MMR keywords.
- 3) **Pain Point Quantification:** Clusters were analyzed based on their frequency (size) and average score. Clusters exhibiting low average scores and high indicators of negative keywords (e.g., 'worse', 'limit', 'broken') were prioritized for detailed discussion as core user pain points.

IV. RESULTS AND ANALYSIS

A. Dataset Characteristics

The final processed dataset consisted of 1,972 valid Reddit posts and comments extracted from the official GPT-5 AMA thread. These posts span from August 7, 2025, to September 29, 2025. Engagement was heavily concentrated immediately after the release announcement, with post scores ranging from 1 to 1,837, indicating strong community participation and diverse sentiment.

B. Clustering Performance Comparison

A systematic evaluation of 12 configurations—two embedding models, two dimensionality reduction methods, and three clustering algorithms—was conducted. The performance was assessed based on Silhouette Score and Calinski-Harabasz Index.

Table I summarizes all clustering outcomes. The UMAP-Nomic + HDBSCAN pipeline demonstrated the best separation and cohesion, achieving:

- **Silhouette Score:** 0.603 (highest across all experiments)
- **21 clusters identified**
- **759 noise points**

TABLE I
CLUSTERING PERFORMANCE RESULTS SUMMARY

Embed. Model	Clust. Algo.	Clusters	Noise	Sil. Score
UMAP-Nomic	HDBSCAN	21	759	0.603
UMAP-Nomic	KMeans	8	0	0.427
UMAP-BGE	HDBSCAN	2	6	0.369
UMAP-BGE	KMeans	8	0	0.368
PCA-BGE	HDBSCAN	2	1659	0.304
PCA-Nomic	HDBSCAN	2	1470	0.202
PCA-Nomic	KMeans	8	0	0.073
PCA-BGE	KMeans	8	0	0.067
UMAP-Nomic	DBSCAN	4	18	-0.059
UMAP-BGE	DBSCAN	5	17	-0.093

To visually verify cluster separation, Figure 1 presents the 2D UMAP projection of all embeddings colored by HDBSCAN labels.

The visualization demonstrates well-separated dense regions corresponding to coherent discussions, while diffuse points form the noise cluster.

C. Topic Modeling Results

BERTopic was applied to the 21 stable clusters identified by HDBSCAN. Because initial c-TF-IDF keywords were dominated by common words, Maximal Marginal Relevance (MMR) re-ranking was used to enhance interpretability. Although Flan-T5 generative labeling encountered instability, MMR-based topic representations were sufficient for meaningful theme identification.

Figure 2 shows the distribution of topic sizes across all identified clusters.

D. Pain Point Identification and Analysis

Four major pain point clusters were identified by combining cluster size, average score, and negative sentiment indicators. Table II presents the detailed statistics.

1) **Removal of Legacy Models:** Clusters 17 and 19 indicate strong dissatisfaction with the removal of GPT-4o. The high score of Cluster 17 reflects widespread demand for reinstating a model selector option.

2) **Creative Writing Quality Degradation:** Cluster 16 highlights user complaints about deteriorated creative writing performance, describing GPT-5 outputs as flat, less imaginative, and less emotionally rich.

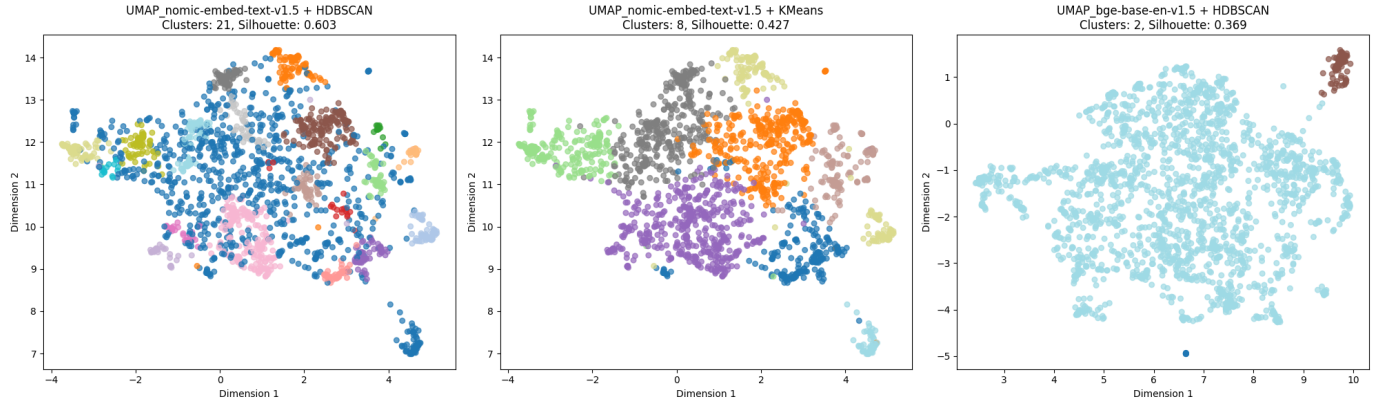


Fig. 1. UMAP visualization of Reddit embeddings clustered using the optimal UMAP–Nomic + HDBSCAN configuration.

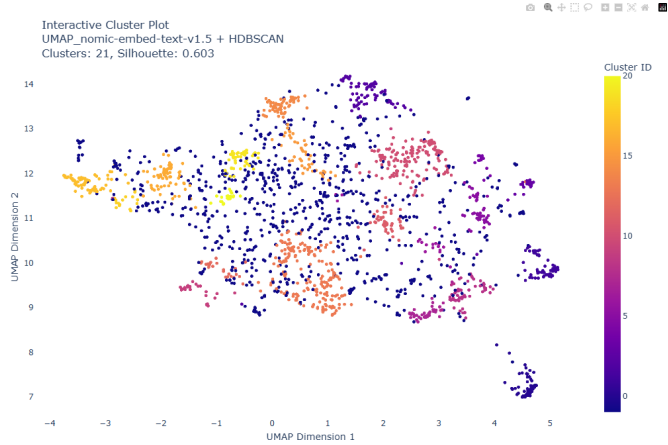


Fig. 2. Topic distribution across the 21 semantic clusters extracted from GPT-5 AMA discussions.

TABLE II
IDENTIFIED PAIN POINT CLUSTERS

ID	Size	Avg Score	Primary Concern	Pain Score
2	93	11.28	Restrictive Usage Limits and Pricing	7
19	47	35.74	Emotional/Personality Downgrade & Legacy Model Loss	5
16	70	20.01	Creative Writing Quality Degradation	0
17	75	34.77	Demand for Legacy Model Selection (GPT-4o)	0

3) *Usage Limitations and Pricing Concerns*: Cluster 2 reveals frustration regarding Plus subscription limits—especially message caps and reduced perceived value relative to competitors.

4) *Personality and Communication Style Changes*: Cluster 19 reflects emotional dissatisfaction with GPT-5’s altered personality. Users describe GPT-5 as colder and less expressive compared to GPT-4o.

V. DISCUSSION

A. Implications for AI Model Deployment

The findings reveal that successful LLM deployment involves not just improving benchmarks, but managing the deep emotional and structural impact on the user base. The strong negative reaction to the removal of GPT-4o highlights that users develop significant model affinity, and abrupt deprecation violates a perceived contract of trust and continuity. Future deployment strategies should consider: (1) maintaining legacy model access for a period, even behind an optional toggle, and (2) prioritizing the maintenance of critical user-facing traits like ‘personality’ and ‘creativity’ alongside technical efficiency. The user backlash over usage limits (Cluster 2) confirms that perceived value erosion, even when coupled with technical upgrades, is a major driver of dissatisfaction and churn risk.

B. Methodological Insights

The empirical success of the **UMAP + HDBSCAN** configuration (Silhouette: 0.603) validates the current best practices in embedding-based text analysis for complex social media data. UMAP’s ability to preserve the local topology of the semantic space, combined with HDBSCAN’s flexibility in identifying varying density clusters and isolating noise, proved superior to traditional methods like PCA/KMeans for discerning nuanced user complaints. Furthermore, the necessity of multi-stage topic modeling (MMR re-ranking) demonstrates that raw clustering output requires refinement to extract truly human-interpretable themes from subjective, emotionally charged text.

C. Broader Implications

The user response to GPT-5 reflects broader tensions in human-AI relationships. Users’ reliance on models for emotional and creative companionship (Cluster 19) suggests that AI systems are moving beyond simple tool utility into a space that requires ethical consideration regarding relational stability and continuity. The demand for user choice in model selection (Cluster 17) suggests that a “one model to rule them all” strategy, while appealing for engineering simplicity, may fail

to meet the diverse and deeply personalized needs of the user community.

VI. CONCLUSION AND FUTURE SCOPE

The study successfully applied advanced NLP techniques to systematically quantify and interpret user pain points arising from the GPT-5 launch. By leveraging high-fidelity embeddings, UMAP for semantic projection, and HDBSCAN for density-based clustering, the methodology provided a rigorous and objective structure for analyzing large volumes of user-generated content. The findings reveal strong user dissatisfaction in three primary areas: the removal of legacy models such as GPT-4o, noticeable reductions in creative and conversational quality, and restrictive usage or pricing limitations for paying subscribers. These insights emphasize that user trust depends not only on technical improvements but also on predictability, emotional continuity, and freedom of choice during major model transitions.

Looking ahead, future work should extend this research through longitudinal analysis to examine how user sentiment evolves over months rather than days. There is also a need for methodological advances that support stable preservation or controlled transfer of “AI personality” across model generations, ensuring emotional and stylistic continuity for long-term users. Additional opportunities include building adaptive interfaces that allow users to switch between model versions based on task requirements, as well as improving generative labeling frameworks with more reliable LLMs to enhance topic interpretability. Together, these directions can contribute to more user-centric and ethically aligned deployment strategies for future LLM releases.